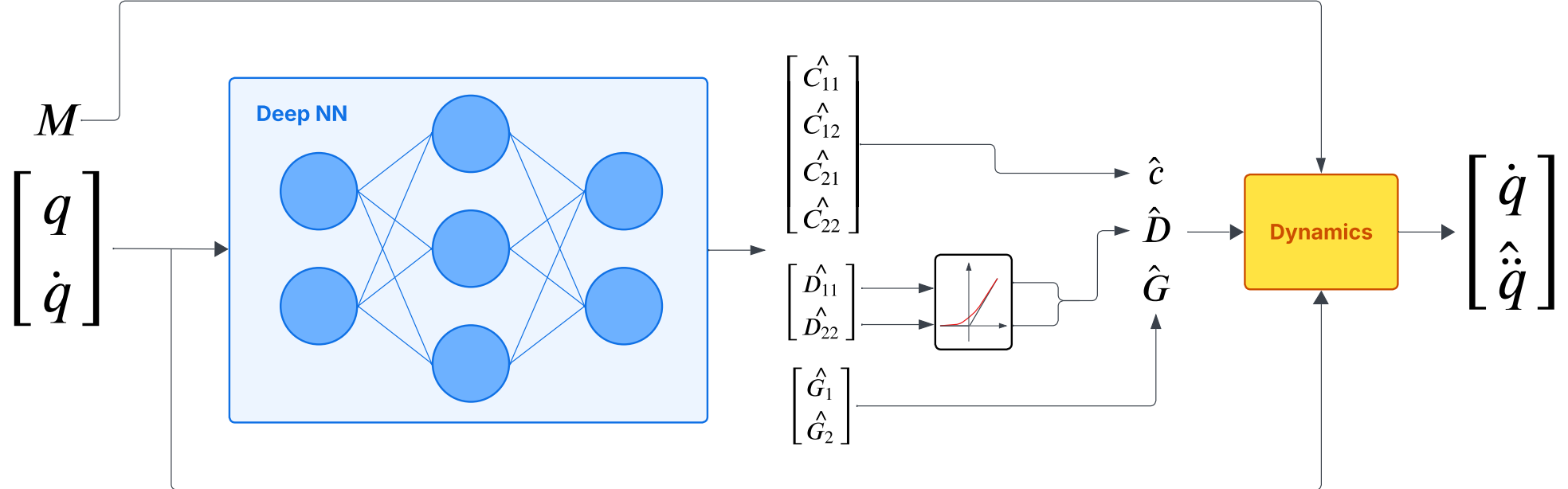
M
$\begin{bmatrix} q \\ \dot{q} \end{bmatrix}$
Deep NN
$\begin{bmatrix} \hat{C}_{11} \\ \hat{C}_{12} \\ \hat{C}_{21} \\ \hat{C}_{22} \end{bmatrix}$
$\begin{bmatrix} \hat{D}_{11} \\ \hat{D}_{22} \end{bmatrix}$
$\begin{bmatrix} \hat{G}_1 \\ \hat{G}_2 \end{bmatrix}$
$\hat{c}$
$\hat{D}$
$\hat{G}$
Dynamics
$\begin{bmatrix} \dot{q} \\ \ddot{\hat{q}} \end{bmatrix}$